IT3010 –Network Design & Management – Labsheet 02

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Batch : Y3.S1.WE.01.02
Date : 18/02/2026

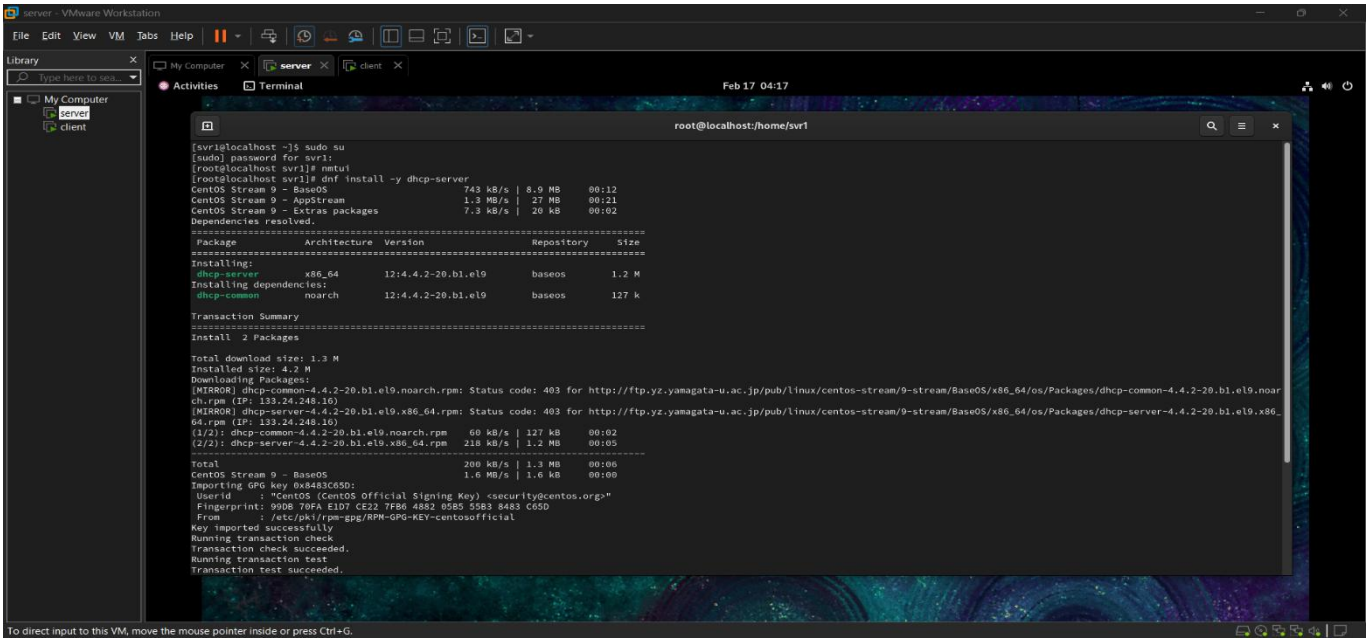
Lab 02 – Installing and Configuring DHCP

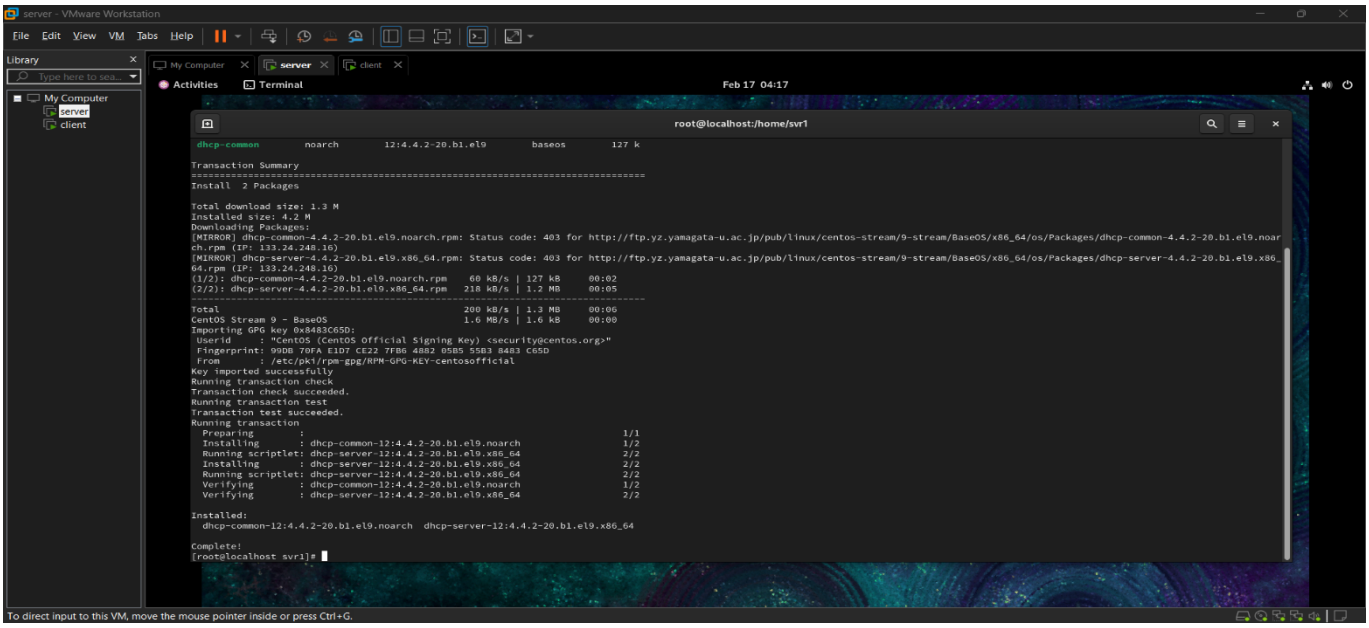
Step 1 – Disable DHCP in VMnet2

- I disabled the DHCP service in VMnet2 from VMware Virtual Network Editor to prevent IP conflicts and ensure that only the CentOS server provides IP addresses.

Step 2 – Install DHCP in CentOS

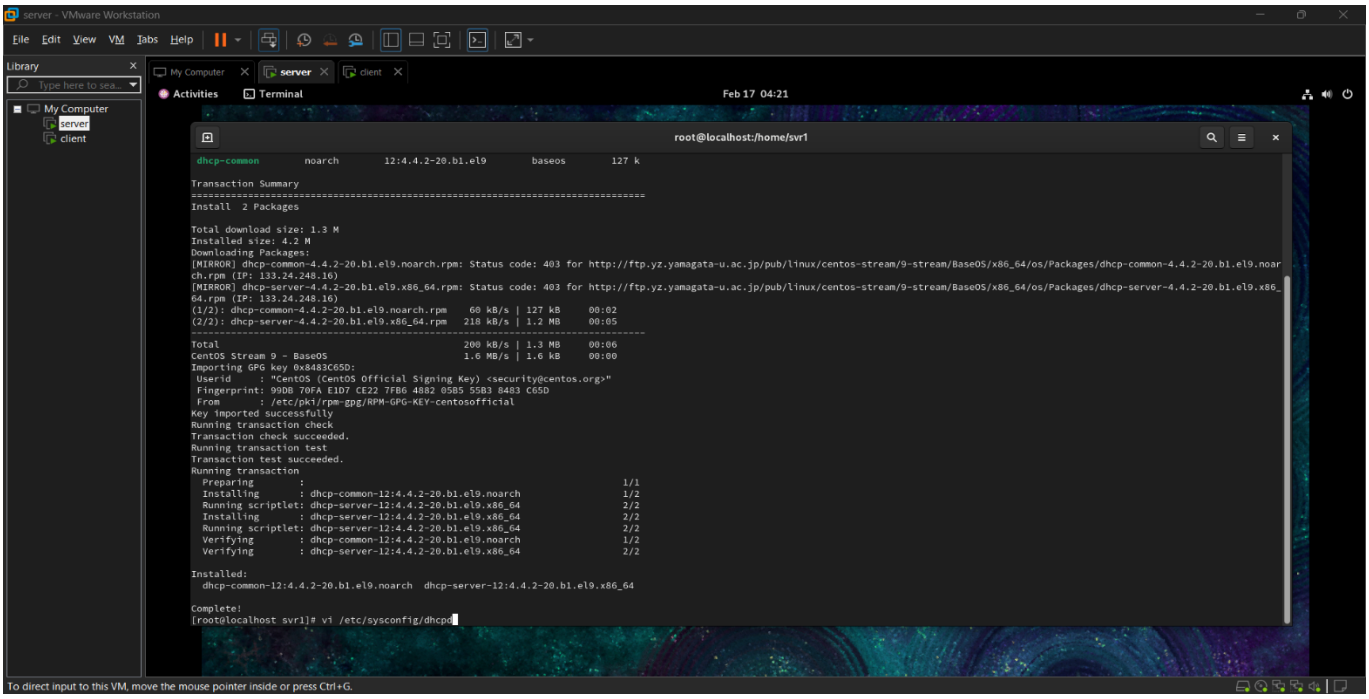
- Command used: dnf install -y dhcp-server
- I installed the DHCP server package in CentOS using the dnf package manager.

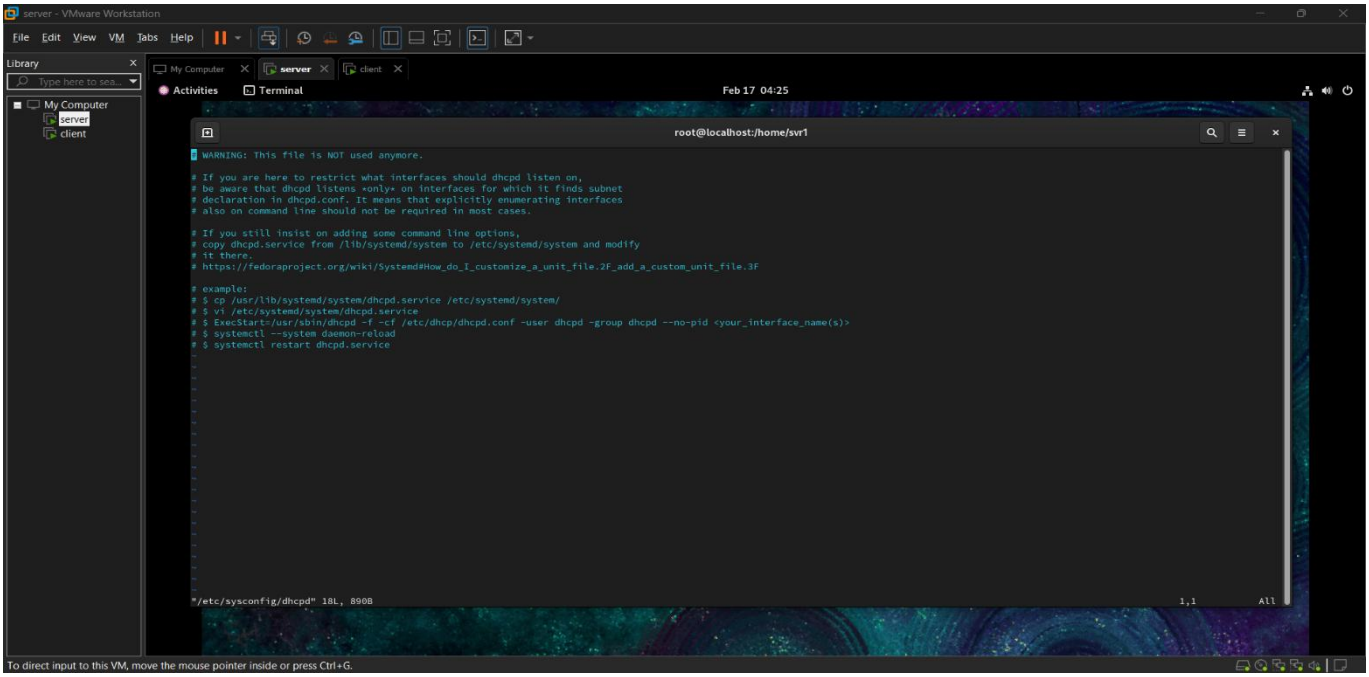


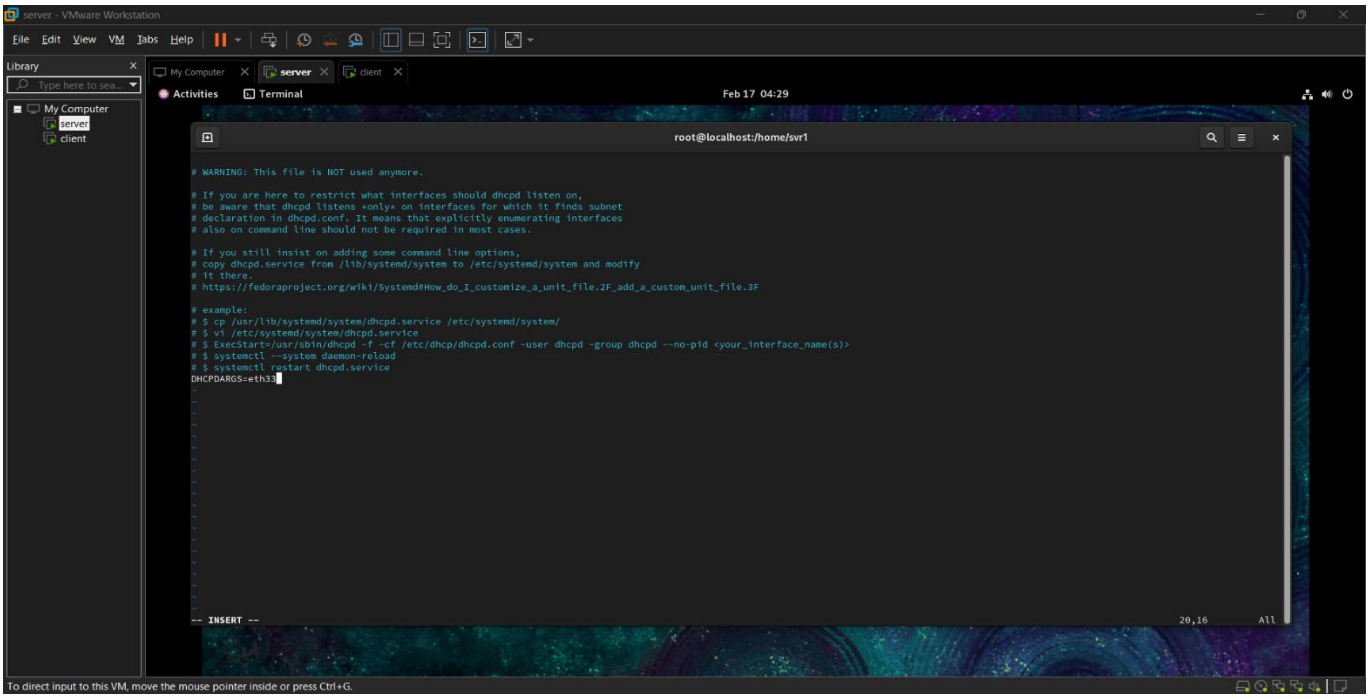


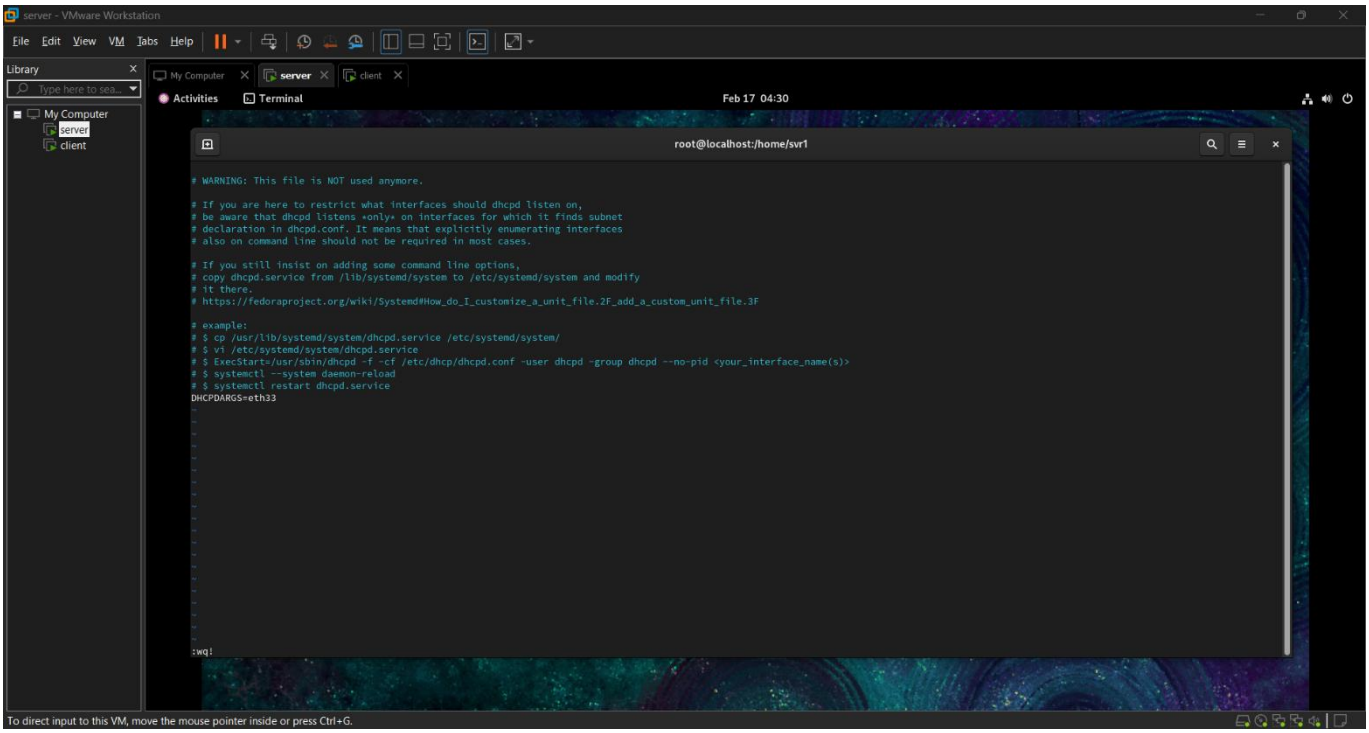
Step 3 – Configure DHCP Interface

- Edited file: vi /etc/sysconfig/dhcpd
- Modified: DHCPDARGS=ens33
- I configured the DHCP server to listen on the correct network interface by editing the /etc/sysconfig/dhcpd file and assigning the interface name.






server - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Activities
Terminal
Feb 17 04:29
root@localhost:/home/svr1
# WARNING: This file is NOT used anymore.
# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.
# If you still insist on adding some command line options,
# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_custom_unit_file.3F
# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid <your_interface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service
DHCPDARGS=eth33
-- INSERT --
20,16
All
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


server - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Activities
Terminal
Feb 17 04:30
root@localhost:/home/svr1
# WARNING: This file is NOT used anymore.
# If you are here to restrict what interfaces should dhcpd listen on,
# be aware that dhcpd listens *only* on interfaces for which it finds subnet
# declaration in dhcpd.conf. It means that explicitly enumerating interfaces
# also on command line should not be required in most cases.
# If you still insist on adding some command line options,
# copy dhcpd.service from /lib/systemd/system to /etc/systemd/system and modify
# it there.
# https://fedoraproject.org/wiki/Systemd#How_do_I_customize_a_unit_file.2F_add_a_custom_unit_file.3F
# example:
# $ cp /usr/lib/systemd/system/dhcpd.service /etc/systemd/system/
# $ vi /etc/systemd/system/dhcpd.service
# $ ExecStart=/usr/sbin/dhcpd -f -cf /etc/dhcp/dhcpd.conf -user dhcpd -group dhcpd --no-pid <your_interface_name(s)>
# $ systemctl --system daemon-reload
# $ systemctl restart dhcpd.service
DHCPDARGS=eth33
:wq!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

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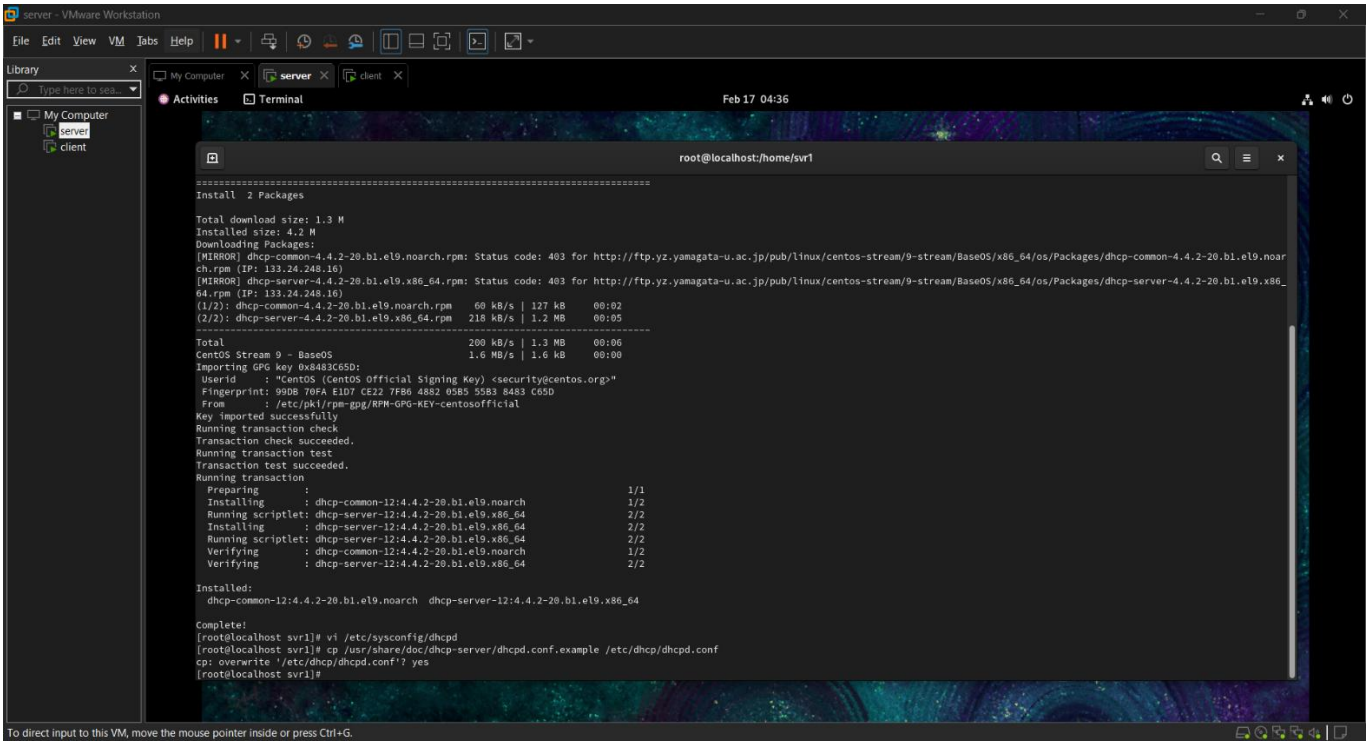
➢ Copy Sample Configuration File

- Command: cp /usr/share/doc/dhcp-server/dhcpd.conf.example /etc/dhcp/dhcpd.conf
- I copied the sample DHCP configuration file to /etc/dhcp/ directory to use it as the main configuration file.



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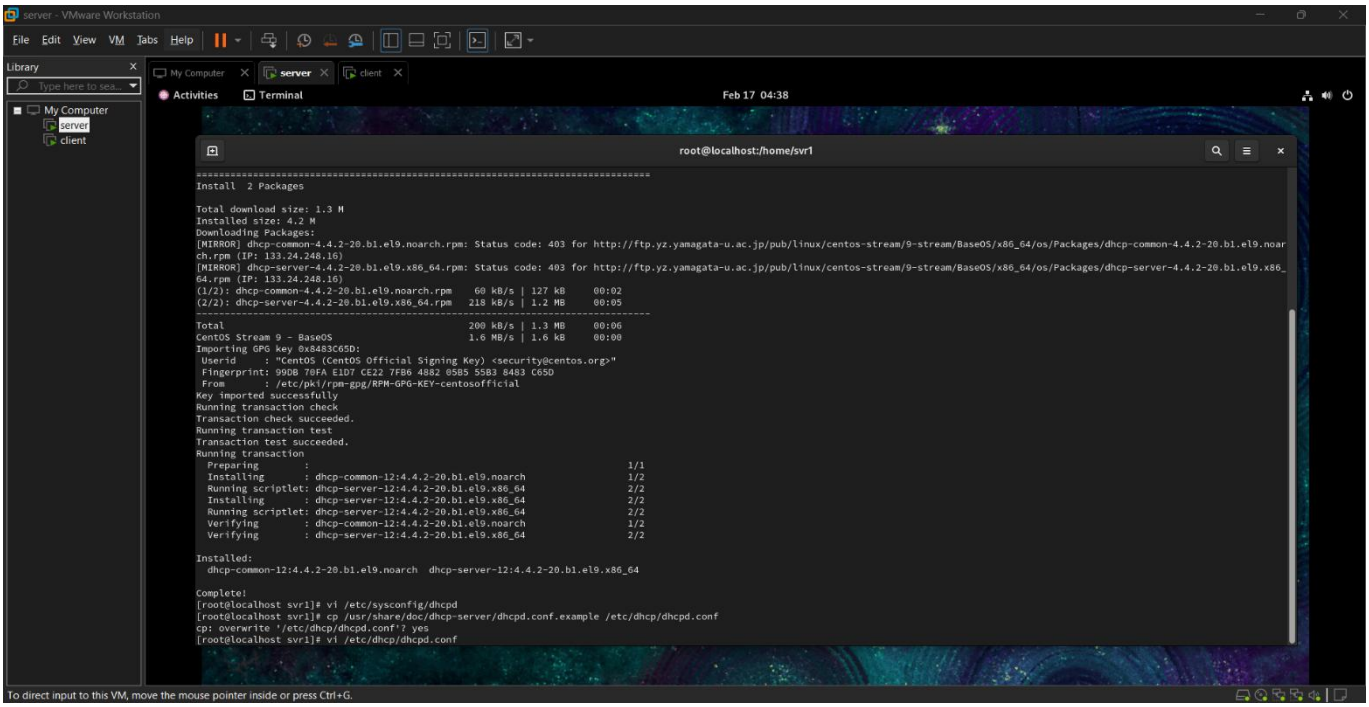
➢ Edit dhcpd.conf File

Command: vi /etc/dhcp/dhcpd.conf

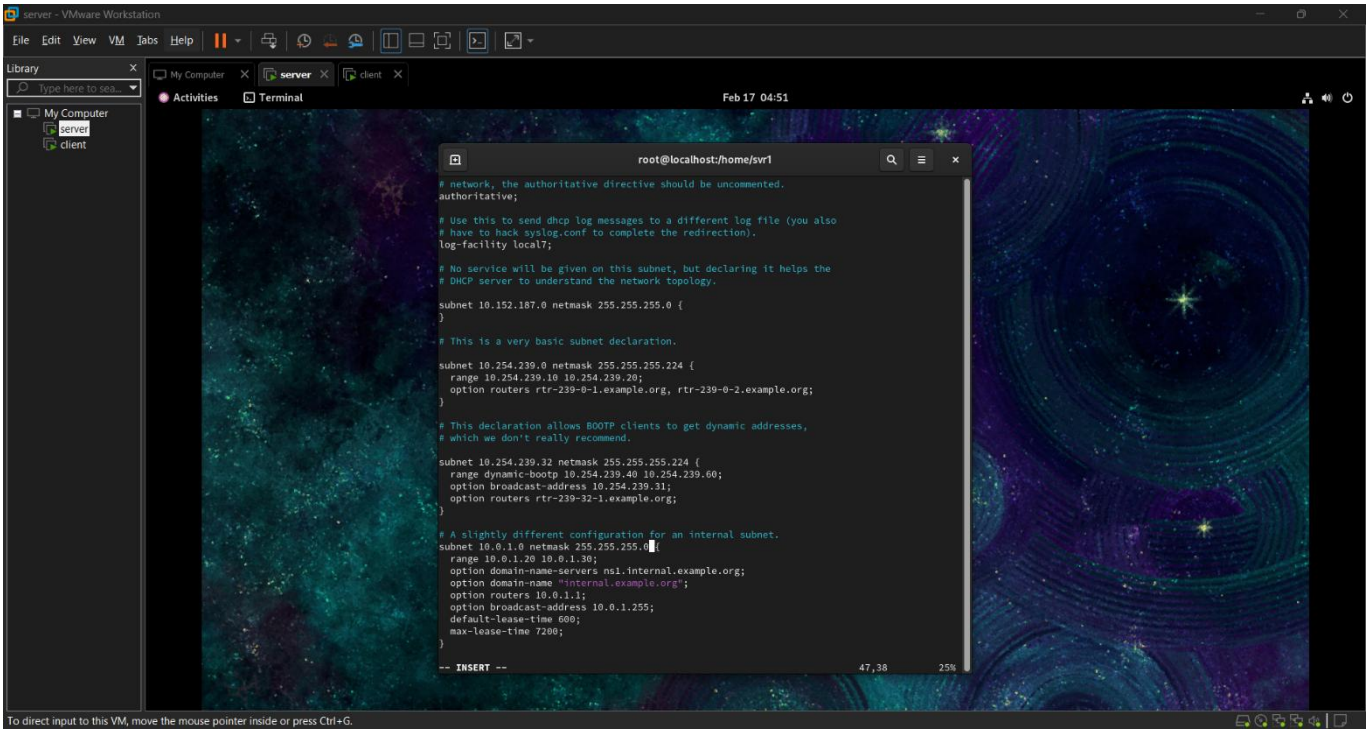
I edited the dhcpd.conf file to define the subnet, IP address range, gateway, broadcast address, lease time, and enabled the authoritative directive.

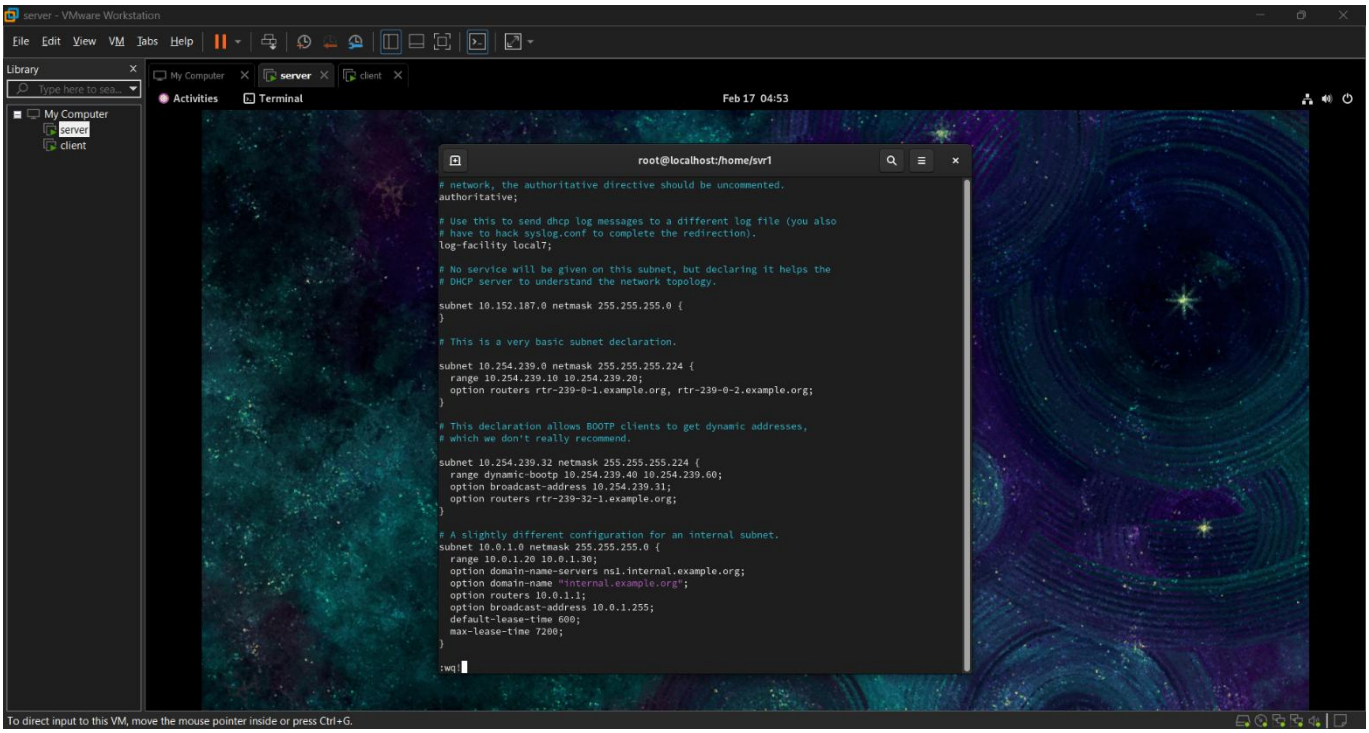


➢ Define Subnet and IP Range

- I configured the DHCP IP pool by defining the subnet 10.0.1.0/24 and assigned an IP range from 10.0.1.20 to 10.0.1.30 for clients.

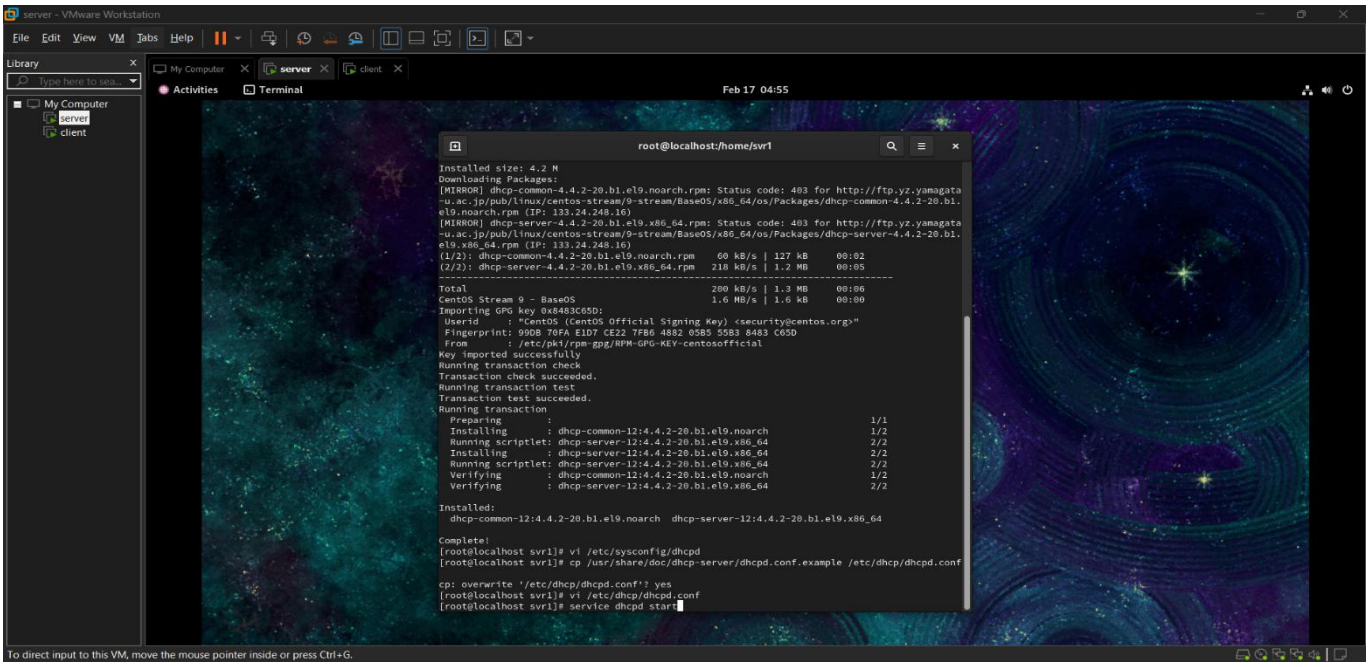
```
subnet 10.0.1.0 netmask 255.255.255.0 {

range 10.0.1.20 10.0.1.30;

option routers 10.0.1.1;

option broadcast-address 10.0.1.255;

default-lease-time 600;

max-lease-time 7200;  }
```


server - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Activities
Terminal
Feb 17 04:51
root@localhost:/home/svr1
network, the authoritative directive should be uncommented.
authoritative;
Use this to send dhcp log messages to a different log file (you also
have to hack syslog.conf to complete the redirection).
log-facility local7;
# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.
subnet 10.152.187.0 netmask 255.255.255.0 {
}
# This is a very basic subnet declaration.
subnet 10.254.239.0 netmask 255.255.255.224 {
range 10.254.239.10 10.254.239.20;
option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}
# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.
subnet 10.254.239.32 netmask 255.255.255.224 {
range dynamic-bootp 10.254.239.40 10.254.239.60;
option broadcast-address 10.254.239.31;
option routers rtr-239-32-1.example.org;
}
# A slightly different configuration for an internal subnet.
subnet 10.0.1.0 netmask 255.255.255.0 {
range 10.0.1.20 10.0.1.30;
option domain-name-servers ns1.internal.example.org;
option domain-name "internal.example.org";
option routers 10.0.1.1;
option broadcast-address 10.0.1.255;
default-lease-time 600;
max-lease-time 7200;
}
-- INSERT --
47,38 25%
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


server - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Activities
Terminal
Feb 17 04:53
root@localhost:/home/svr1
network, the authoritative directive should be uncommented.
authoritative;
Use this to send dhcp log messages to a different log file (you also
have to hack syslog.conf to complete the redirection).
log-facility local7;
# No service will be given on this subnet, but declaring it helps the
# DHCP server to understand the network topology.
subnet 10.152.187.0 netmask 255.255.255.0 {
}
# This is a very basic subnet declaration.
subnet 10.254.239.0 netmask 255.255.255.224 {
range 10.254.239.10 10.254.239.20;
option routers rtr-239-0-1.example.org, rtr-239-0-2.example.org;
}
# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.
subnet 10.254.239.32 netmask 255.255.255.224 {
range dynamic-bootp 10.254.239.40 10.254.239.60;
option broadcast-address 10.254.239.31;
option routers rtr-239-32-1.example.org;
}
# A slightly different configuration for an internal subnet.
subnet 10.0.1.0 netmask 255.255.255.0 {
range 10.0.1.20 10.0.1.30;
option domain-name-servers ns1.internal.example.org;
option domain-name "internal.example.org";
option routers 10.0.1.1;
option broadcast-address 10.0.1.255;
default-lease-time 600;
max-lease-time 7200;
}
:wq!
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

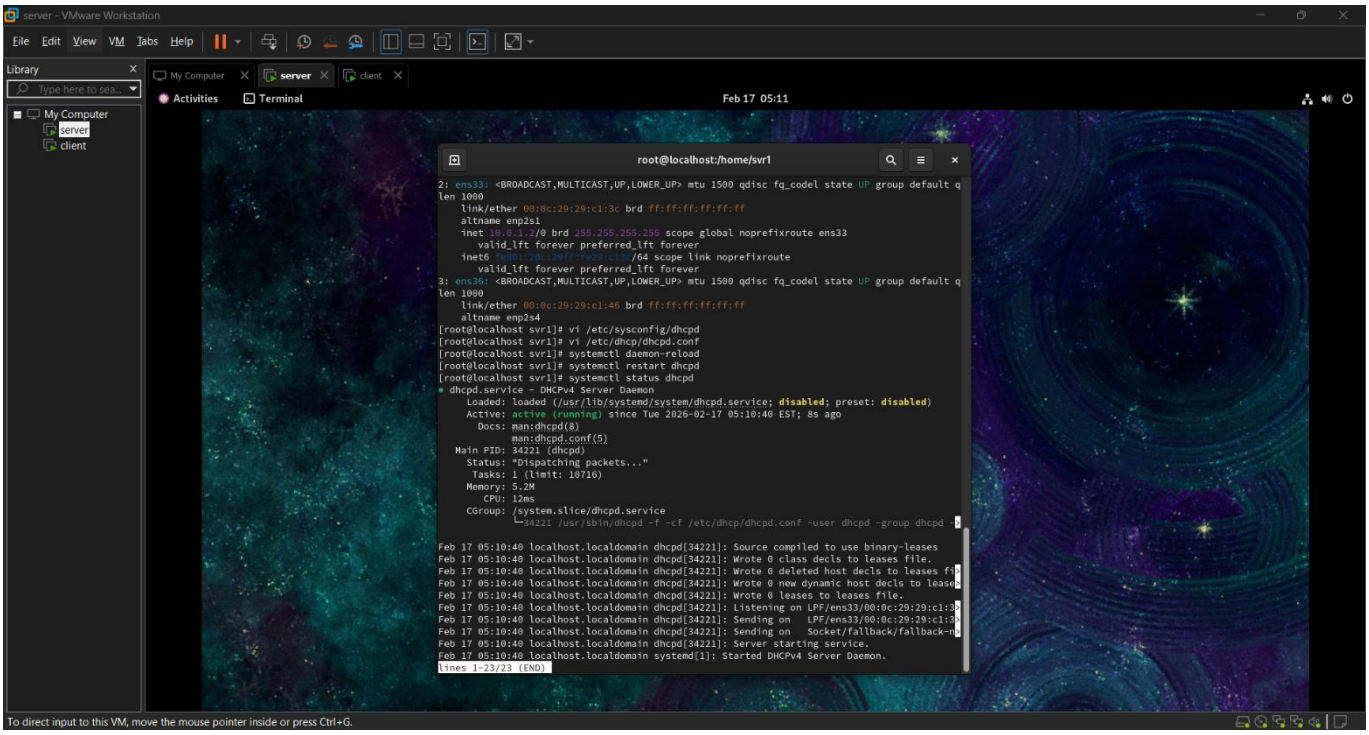
➢ Start DHCP Service

- Command: service dhcpd start
- I started the DHCP service to activate the DHCP server.



➢ Check DHCP Status

- Command: systemctl status dhcpd
- I verified that the DHCP service is active and running successfully using the systemctl command.



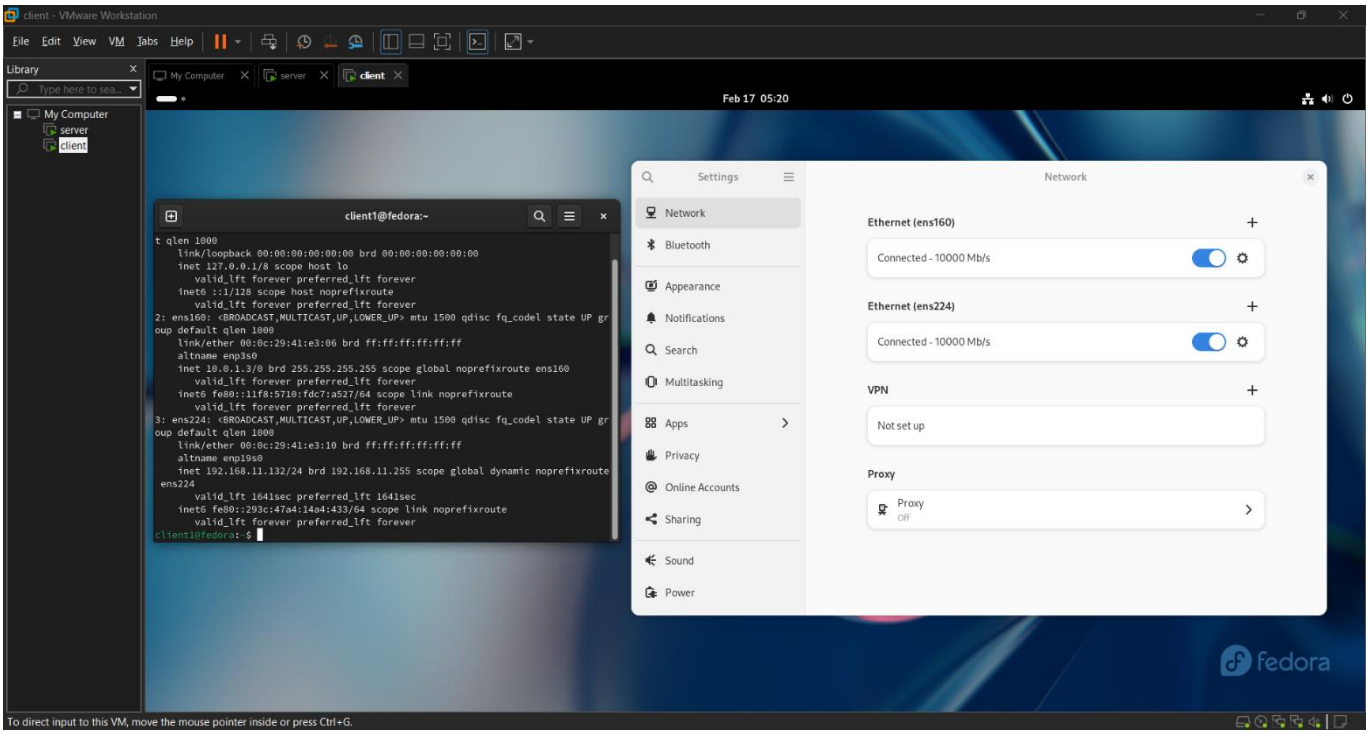
➢ Configure Client to Obtain IP Automatically

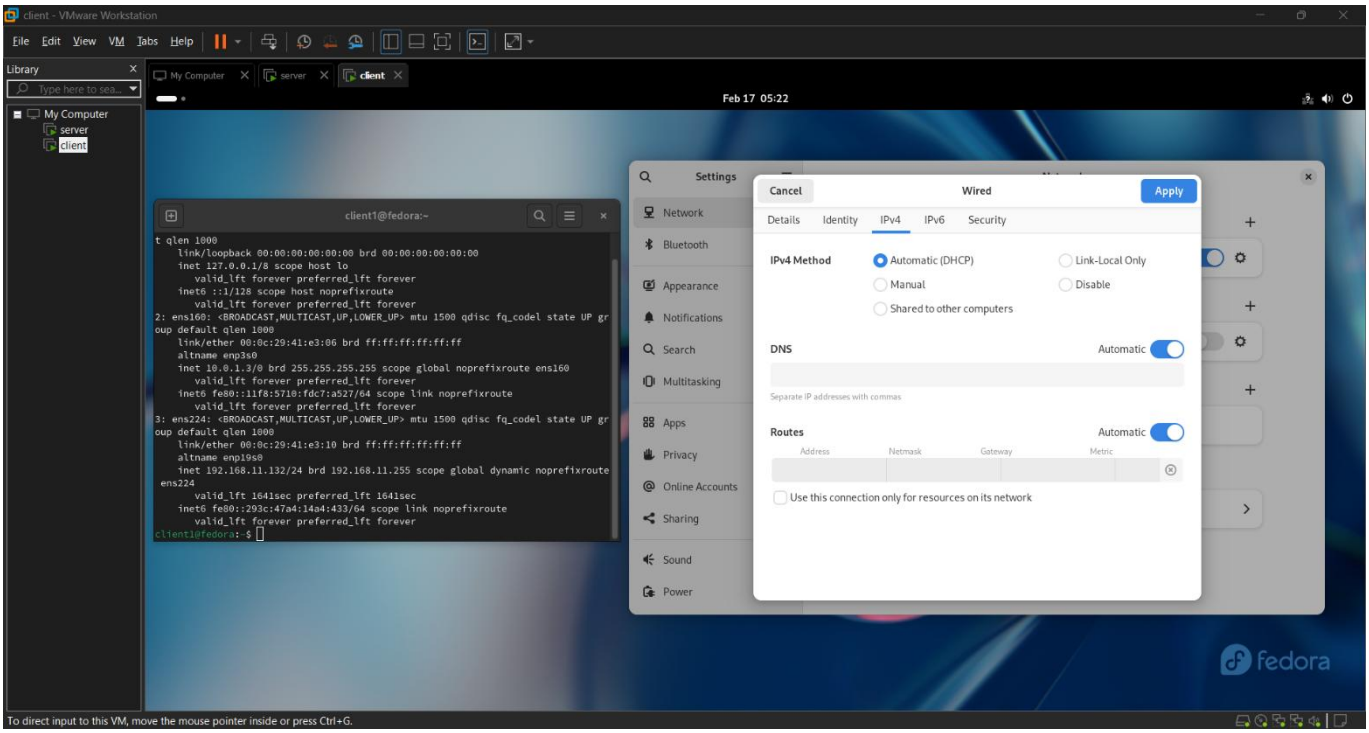
On Fedora client:

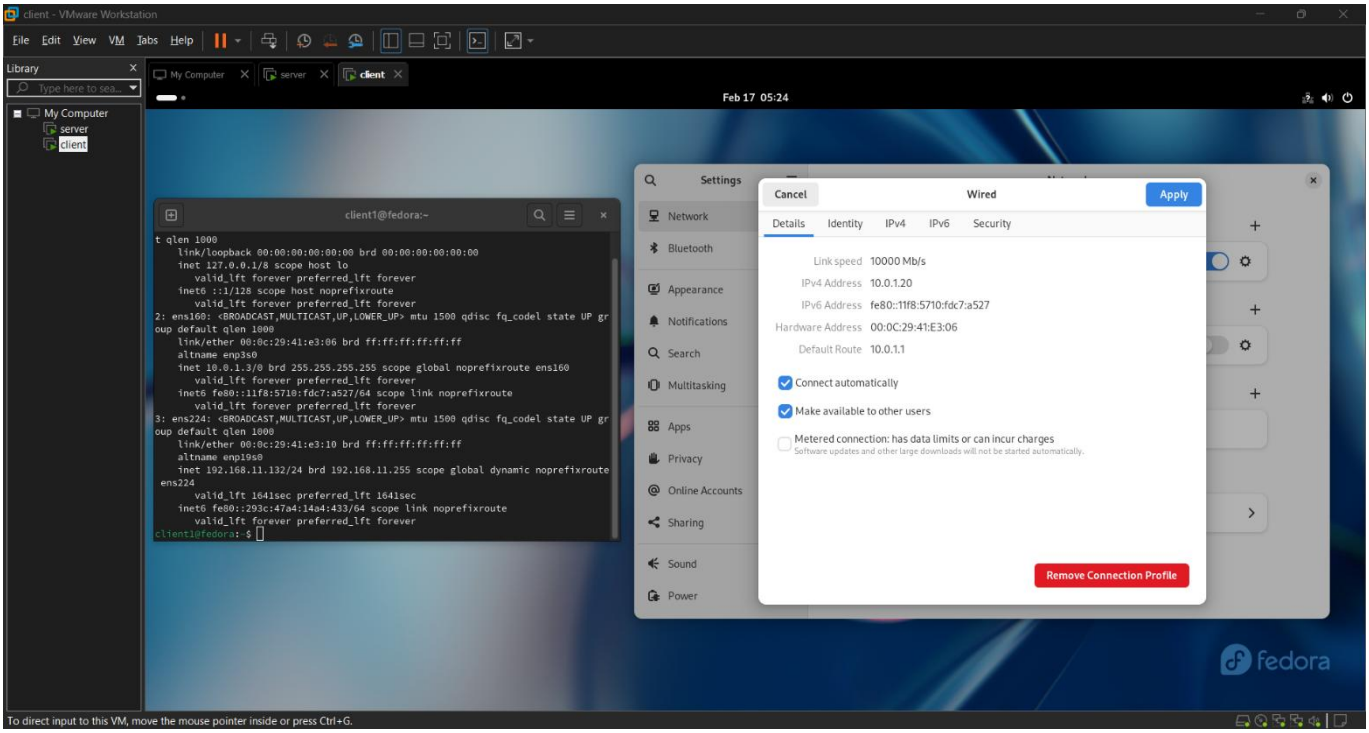
- Open Network Settings
- Set IPv4 Method to Automatic (DHCP)

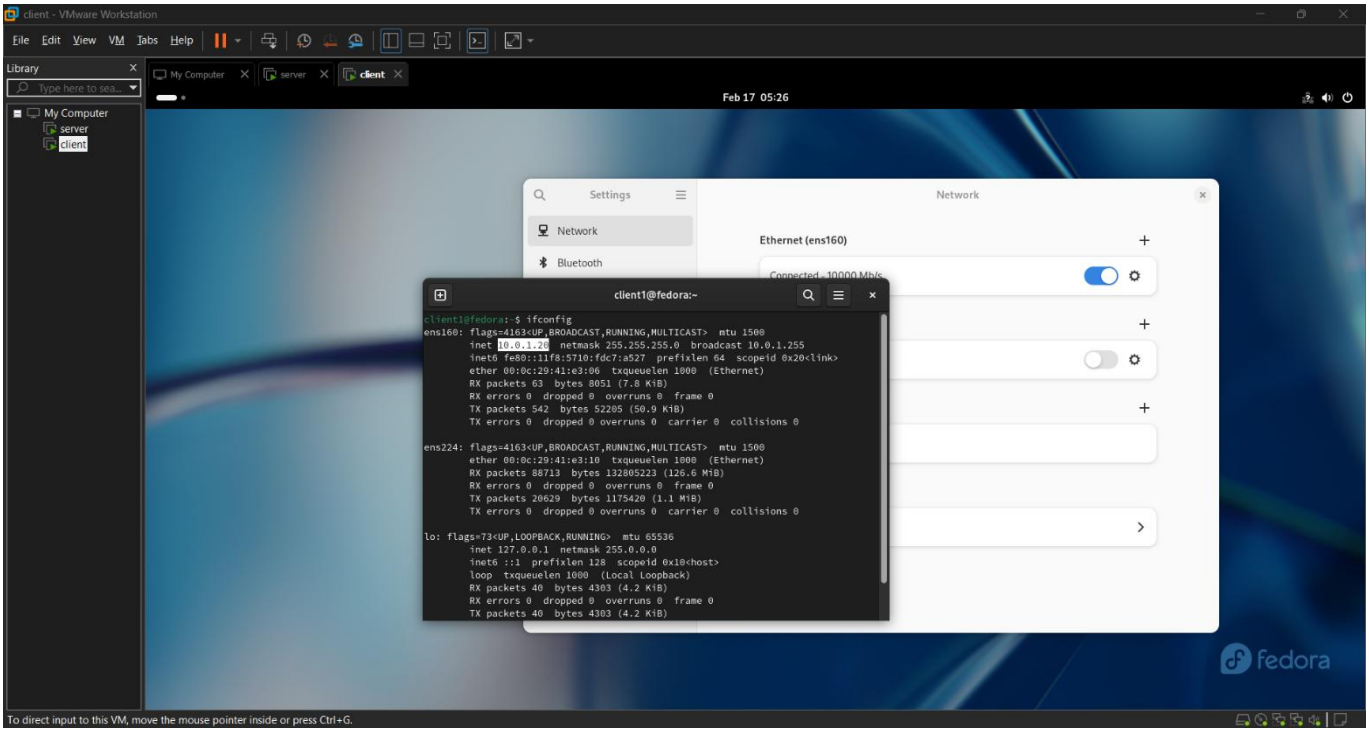
What I did:

I changed the Fedora client's IPv4 configuration to Automatic (DHCP) so that it can obtain an IP address from the DHCP server.




client - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Feb 17 05:22
client1@fedora:~
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:41:e3:06 brd ff:ff:ff:ff:ff:ff
altname enp3s0
inet 10.0.1.3/0 brd 255.255.255.255 scope global noprefixroute ens160
valid_lft forever preferred_lft forever
inet6 fe80::11f8:5710:fdc7:a527/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:41:e3:10 brd ff:ff:ff:ff:ff:ff
altname enp19s8
inet 192.168.11.132/24 brd 192.168.11.255 scope global dynamic noprefixroute ens224
valid_lft 1641sec preferred_lft 1641sec
inet6 fe80::293c:47a4:14a4:433/64 scope link noprefixroute
valid_lft forever preferred_lft forever
client1@fedora:~$
Settings
Network
Bluetooth
Appearance
Notifications
Search
Multitasking
Apps
Privacy
Online Accounts
Sharing
Sound
Power
Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
IPv4 Method
Automatic (DHCP)
Link-Local Only
Manual
Disable
Shared to other computers
DNS
Automatic
Separate IP addresses with commas
Routes
Automatic
Address
Netmask
Gateway
Metric
Use this connection only for resources on its network
fedora
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


client - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
server
client
Feb 17 05:24
client1@fedora:~
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:41:e3:06 brd ff:ff:ff:ff:ff:ff
altname enp3s0
inet 10.0.1.3/0 brd 255.255.255.255 scope global noprefixroute ens160
valid_lft forever preferred_lft forever
inet6 fe80::11f8:5710:fdc7:a527/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: ens224: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:41:e3:10 brd ff:ff:ff:ff:ff:ff
altname enp19s8
inet 192.168.11.132/24 brd 192.168.11.255 scope global dynamic noprefixroute ens224
valid_lft 1641sec preferred_lft 1641sec
inet6 fe80::293c:47a4:14a4:433/64 scope link noprefixroute
valid_lft forever preferred_lft forever
client1@fedora:~$
Settings
Network
Bluetooth
Appearance
Notifications
Search
Multitasking
Apps
Privacy
Online Accounts
Sharing
Sound
Power
Cancel
Wired
Apply
Details
Identity
IPv4
IPv6
Security
Link speed 10000 Mb/s
IPv4 Address 10.0.1.20
IPv6 Address fe80::11f8:5710:fdc7:a527
Hardware Address 00:0C:29:41:E3:06
Default Route 10.0.1.1
Connect automatically
Make available to other users
Metered connection: has data limits or can incur charges
Software updates and other large downloads will not be started automatically.
Remove Connection Profile
fedora
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

➢ Verify IP Address on Client

- Command: ifconfig
- I checked the client's IP address using the ifconfig command to confirm that it received an IP from the configured DHCP range.